THE INFLUENCE OF TV SHOWS ON BABY NAME TRENDS IN CANADA

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1. INTRODUCTION

I came across Reddit and blog posts where people are considering naming their babies "Luffy" after the release of the live-action *One Piece* adaptation. This made me curious: how much do popular TV shows influence baby name trends in Canada? Do people really name their children after characters from popular TV shows? In this project, I imagined a scenario where a parent wants to name their baby after their favorite character from a popular TV show that just came out this year but wonders whether others might do the same, and if the name is likely to become popular. To explore this, I conducted a study to answer the question: **Do popular TV show characters influence baby name usage in Canada?** To quantify the popularity, I defined "popular" shows as those with high IMDb ratings.

2. DATASET

This section describes the sources of the data and steps taken to clean and prepare the data for analysis.

2.1: COLLECTING RAW DATA AND SOURCES

I obtained 4 individual datasets:

1. `Baby-names-frequency_1980_2020.xlsx:` An Excel file containing baby names registered in Canada from 1980 to 2020, including the year of birth, gender, and the number of occurrences (frequency).
2. `Baby-names-frequency_2024.xlsx:` An Excel file with baby name data from 2021 to 2024, using the same structure as the previous dataset.
3. `IMDB.csv:` A CSV file containing metadata for movies and TV shows, including unique IDs, title names, release years, average ratings, languages, and regions. It includes media data starting from the year 1990.
4. `Title.principals.tsv:` A TSV file with information about key cast and crew for each title, including unique IDs, role types, and character names.

The first two are from the Government of Canada, and the latter two are from the IMDb Developer.

2.2: BABY NAME DATASET IN CANADA

From the baby-name frequency Excel files, I extracted the following columns:

- First name
- Year of birth (ranging from 1987 to 2024)
- Frequency (number of occurrences)

I kept baby name data from 1987 onward and excluded data from 1980 to 1986 because the IMDb dataset only includes shows released in 1990 or later. To create a ±3-year analysis window around each show's release year, I needed baby name data starting at least three years before 1990, which is 1987. Therefore, data before 1987 would not be used in the analysis. I combined the two files and converted the data into a single CSV file named `baby_names_by_year.csv` with the columns: `[name, year, count]`.

2.3: DATASET OF MAIN CHARACTERS FROM POPULAR TV SHOWS

Since no single dataset contained TV show titles, release year, and main character names, I created one by combining several datasets. This process involved two main steps:

2.3.1: FILTERING POPULAR SHOWS

I applied the following filters to identify popular shows from `IMDB.csv`:

- **English-language TV shows**: To match the linguistic context of Canadian baby name data
- **With at least 5,000 reviews**: To filter out less-viewed shows and focus on those with a large enough audience
- **An average rating of 8.0 or higher**: To include only well-received shows
- **Released between 1990 and 2021**: The `IMDB.csv` file includes TV shows released between 1990 and 2024. To maintain a full ±3-year analysis window using baby name data (which is available up to 2024), we included only shows released up to

and including 2021. Shows released after 2021 were excluded because we would not have baby name data for the full three years following their release.

The resulting dataset, `popular_shows.csv`, contains the following columns:

- `tconst`: Unique IMDb ID for each show
- `primaryTitle`: Title of the show
- `startYear`: Year the show premiered

2.3.2: EXTRACTING MAIN CHARACTER NAMES

Since `IMDB.csv` did not include character names, I used a second dataset, `title.principals.tsv`, which includes:

- `tconst`: Unique IMDb ID
- `ordering`: Indicates the character's prominence (1 = lead role)
- `job`: Role type (e.g., actor, actress, director)
- `characters`: Name of the character played

I filtered this file to include:

- Only the shows listed in `popular_shows.csv`
- Only actors and actresses with `ordering` values of 1 to 3 (i.e., the top three main characters)

After filtering, I created a dataset called `main_characters`, which includes `tconst` and `character_names`.
I then joined it with the `popular_shows` dataset from section 2.3.1 using the `tconst` as the key.The resulting dataset, `main_characters.csv`, includes the following columns: `[tconst, characters, primaryTitle, startYear]`.

2.4: CLEANING FIRST NAMES

`Characters` in `main_characters` were full names enclosed in brackets (e.g., ["FirstName LastName"]). Also, some included common prefixes like “Mr.” or “Detective” while others had uncommon ones such as “godmother” or “samurai”. There were also invalid names such as “Narrator” or “Self”. To clean the data, I manually reviewed the character names and identified both common and uncommon prefixes. Then, I removed these prefixes, stripped brackets, and filtered out invalid entries.

2.5: COMBINE CANADIAN BABY NAME DATA WITH CLEANED MAIN CHARACTER NAMES

After cleaning the `main_characters`, I merged them with the `baby_names_by_year.csv.` For each `(firstName, primaryTitle)` pair, I created a 7-year window around the show’s release year (±3 years). For each year in this window:

- If a name did not appear in `baby_names_by_year.csv`, its count was set to **0**
- Otherwise, I calculated a `years_from_release` value:
  `years_from_release = baby_name_year - show_release_year`

I removed `(firstName, primaryTitle)` pairs with fewer than 30 total occurrences in the 7-year window because rare names are more likely to reflect noise rather than real trends. The final dataset, `name_trends.csv`, includes 4,550 rows with the following structure:

firstName	title	years_from_release	count
James	In Living Color	-3	259
James	In Living Color	-2	239
James	In Living Color	-1	275
James	In Living Color	0	239
James	In Living Color	1	265
James	In Living Color	2	230
James	In Living Color	3	189
Jake	In Living Color	-3	10
Jake	In Living Color	-2	13

3. TECHNIQUES USED TO ANALYZE THE DATA

This section describes how I checked the assumptions required for a t-test, and how I used the test to determine whether there was a significant difference in baby name usage before and after a show's release.

3.1: RATIONALE FOR USING THE T-TEST

I selected the t-test because the objective was to compare two independent groups: pre-release and post-release name counts. This method is appropriate for detecting differences in group means. As described below, the necessary assumptions were met after applying appropriate data transformations.

3.2: CHECKING ASSUMPTIONS

Before conducting the t-test, I needed to verify that the required assumptions (normality, equal variance, and independence) were met. I split the data as follows:

- **Pre-release:** `years_from_release` between -3 and -1 (inclusive)
- **Post-release:** `years_from_release` between 0 and 3 (inclusive)

3.2.1 NORMAL DISTRIBUTION:

Figure 1 shows the distribution of the raw data, which is highly right-skewed. The distribution deviated significantly from normality, so the Central Limit Theorem could not be applied. Therefore, I applied both square root and log transformations to improve normality. While normality tests still failed after both transformations, the log-transformed data appeared close enough to normal to proceed with the t-test (see Figure 2).

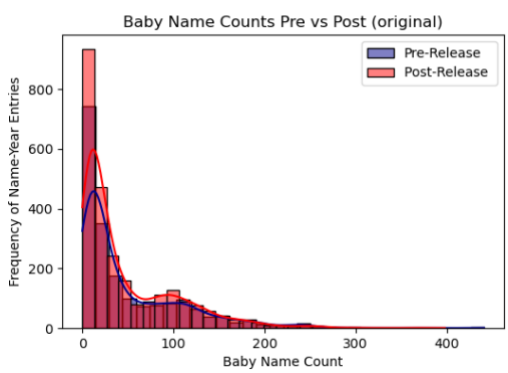

Figure 1: Data distribution without transformation

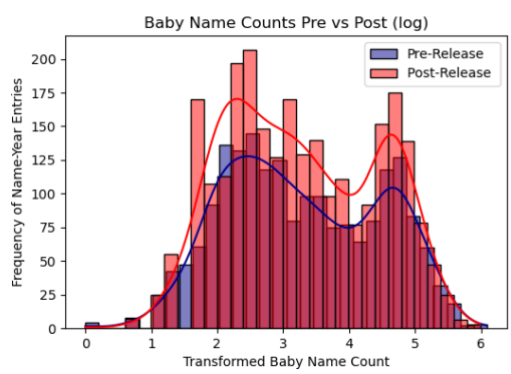

Figure 2: Data distribution with log transformation

3.2.2: EQUAL VARIANCE

I used Levene's test to assess whether the variances of the two groups (pre- and post-release) were equal based on the log-transformed data.

- **Null hypothesis (H_0):** The variances are equal
- **Alternative hypothesis (H_1):** The variances are not equal
- **p-value:** 0.201

Since the p-value is greater than 0.05, I failed to reject the null hypothesis and concluded that the assumption of equal variance is reasonably satisfied.

3.2.3: INDEPENDENCE AND IDENTICAL DISTRIBUTION

The baby name data is grouped by year, and each year represents a separate observation. Because the pre- and post-release periods do not overlap, there is no dependency between the two groups. Therefore, the independence assumption is considered valid. Additionally, name count data are collected over a fixed number of years using the same methodology. Thus, the identical distribution assumption is considered valid.

3.3: CONDUCTING T-TEST

Since all 3 assumptions were reasonably satisfied after the log transformation, I proceeded with the t-test using the ttest_ind function.

- **Null hypothesis (H_0):** $\mu_1 = \mu_2$
- **Alternative hypothesis (H_1):** $\mu_1 \neq \mu_2$

Where:

- μ_1 = mean name count in the pre-release period
- μ_2 = mean name count in the post-release period

3.4: USING THE MANN-WHITNEY U TEST TO DETECT SHOW-SPECIFIC TRENDS

3.4.1: PURPOSE OF TESTING INDIVIDUAL PAIRS

I also performed the Mann-Whitney U test to identify specific (name, show) pairs that showed increasing or decreasing trends after a show's release. I chose the Mann-Whitney U test because many of these pairs had non-normal distribution even after transformation and did not meet the assumptions required for a t-test.

3.4.2: HYPOTHESES

Let:

- μ_1 = mean name count in the pre-release period
- μ_2 = mean name count in the post-release period

For testing an increasing trend in name usage after the show's release:

- **Null hypothesis (H_0):** $\mu_1 \geq \mu_2$
- **Alternative hypothesis (H_1):** $\mu_1 < \mu_2$

For testing a decreasing trend:

- **Null hypothesis (H_0):** $\mu_1 \leq \mu_2$
- **Alternative hypothesis (H_1):** $\mu_1 > \mu_2$

3.4.3: JUSTIFICATION FOR ONE-TAILED TEST

I used a one-tailed test because I was specifically interested in the direction of the change. For a decreasing trend, I only cared if post-release name counts were significantly lower than pre-release counts. I was not interested in whether they were equal or higher. Similarly, for an increasing trend, I only tested whether post-release counts were significantly higher and was not interested in cases where they were equal or lower.

4. RESULTS

This section presents the results of the t-test and Mann-Whitney U test conducted in Section 3.

4.1: RESULTS OF T-TEST

The resulting p-value: **0.6616**

Although figure 3 displays a somewhat decreasing trend, I failed to reject the null hypothesis. In other words, **<u>we do not have enough evidence to conclude that there is a statistically significant difference in name usage between the pre- and post-release periods.</u>**

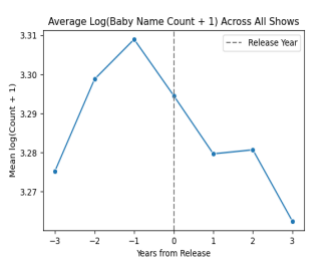

Figure 3: the average log-transformed baby name counts across all shows

4.2: RESULTS OF MANN-WHITNEY U TEST

A total of **56 TV shows were associated with a significant increase** in baby name usage after their release.(e.g., *Eleanor* from *Black Sails*, *Veronica* from *Veronica Mars*). On the other hand, **96 shows were linked to a significant decrease** (e.g.,Annalise from *How to Get Away with Murder* and *Anna* from both *Amphibia* and *PEN15*). These patterns are shown in Figure 4 and Figure 5, which display the top 5 shows with the most significant increases and decreases in baby name usage. Note that the complete results are available in the `results/` directory of the Git repository.

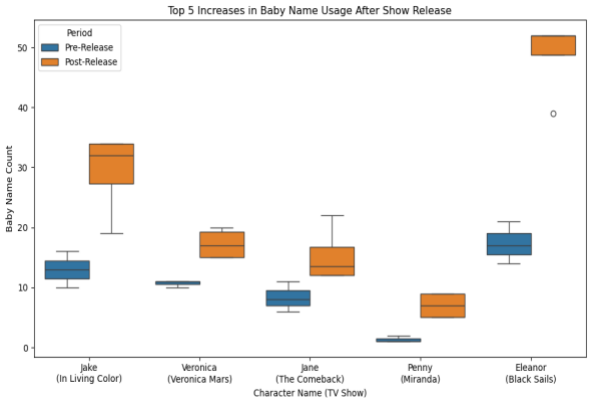

Figure 4: the top 5 shows with significant increasing baby name usage after release

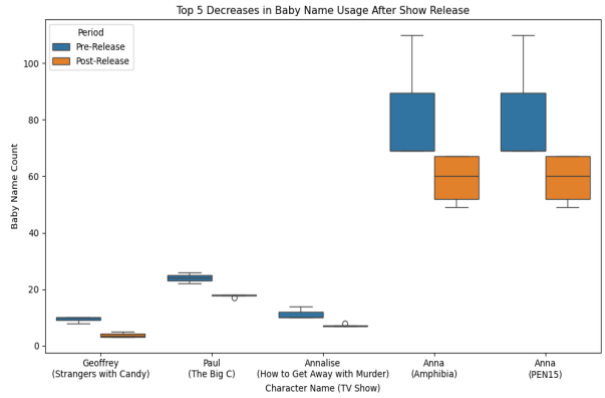

Figure 5: the top 5 shows with significant decreases in baby name usage after release

5. LIMITATIONS AND CHALLENGES

In this section, I present the challenges I encountered throughout the process and acknowledge the limitations of my analysis and testing.

5.1: PROBLEMS ENCOUNTERED

Cleaning Character Names:
This step was challenging because I had to manually inspect the character names to identify and remove various prefixes and noise. The formatting varied widely, and it required careful review.

Merging Baby Name and TV Show Data:
While reviewing the result summary, I noticed that some common baby names had a count of 0 for certain (name, show) pairs (e.g., *William* from *Stage*). This clearly indicated issues in the data cleaning and merging process and it was due to inconsistencies in the baby name Excel files. Some entries contained unexpected leading spaces or empty strings, causing identical names to be treated as different during processing. It took several hours to identify and correct the issue.

5.2: LIMITATIONS OF TEST

- Other factors may influence:
Baby name trends can also be affected by factors such as celebrity influence, immigration, and cultural events. Therefore, even for names with the most significant increases or decreases, we cannot conclude that the TV show was the sole cause of the change.

- Character Likability Not Considered:
Not all characters are likable or name-worthy, which could influence naming decisions. Unfortunately, I was unable to find any dataset that rated character likability, which limited the accuracy of the trend analysis.

- Restricted Time Window (±3 Years):
The analysis was limited to a ±3-year window around each show's release due to constraints in both the TV show and baby name data. I believe the analysis could have captured longer-term trends more accurately with a broader time range.